

Data, Programming, Cloud

Don Syme, @dsyme

Microsoft Research, F# Community Contributor

Our Expertise (at MSR Cambridge)

Programming!

More Productive Programming!

Safer Programming!

More Correct Programming!

More Interoperable Programming!

More Efficient Programming!

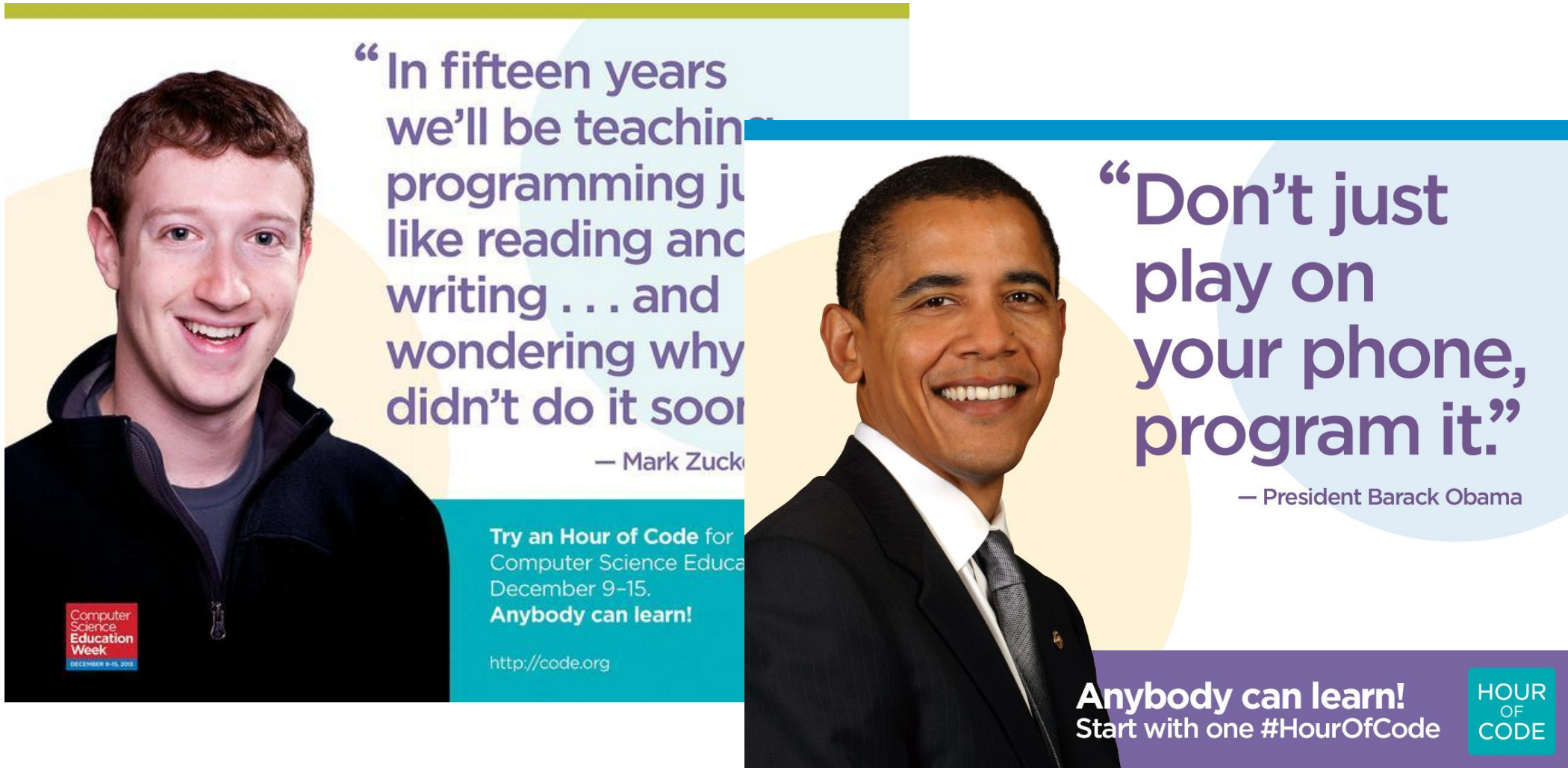
More Enjoyable Programming!

....

← Lots of ideas needed!

← Lots of technical work!

Programming matters!



The graphic is split into two main sections. The left section features a portrait of Mark Zuckerberg on a yellow circular background. To his right, a light blue box contains a quote in purple text. Below the quote, a teal box contains promotional text for Computer Science Education Week and the Hour of Code. The right section features a portrait of Barack Obama on a yellow circular background. To his right, a light blue box contains a quote in purple text. Below the quote, a purple box contains promotional text for the Hour of Code, including a hashtag and a logo.

“In fifteen years we’ll be teaching programming just like reading and writing . . . and wondering why didn’t do it sooner.”
— Mark Zuckerberg

Try an Hour of Code for Computer Science Education Week
December 9–15.
Anybody can learn!
<http://code.org>

“Don’t just play on your phone, program it.”
— President Barack Obama

Anybody can learn!
Start with one #HourOfCode

HOUR OF CODE

Programming matters!

The worldwide view:

10Million devs x \$100K/year productivity

=

\$1Trillion/year

1% productivity increase = \$1Billion!!

But isn't today is about data and
cloud!?!?

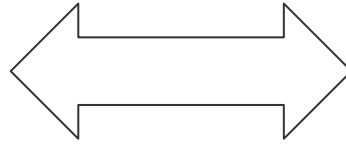
X



Y

Code

In one large
company, ~500
people work on
tooling for this



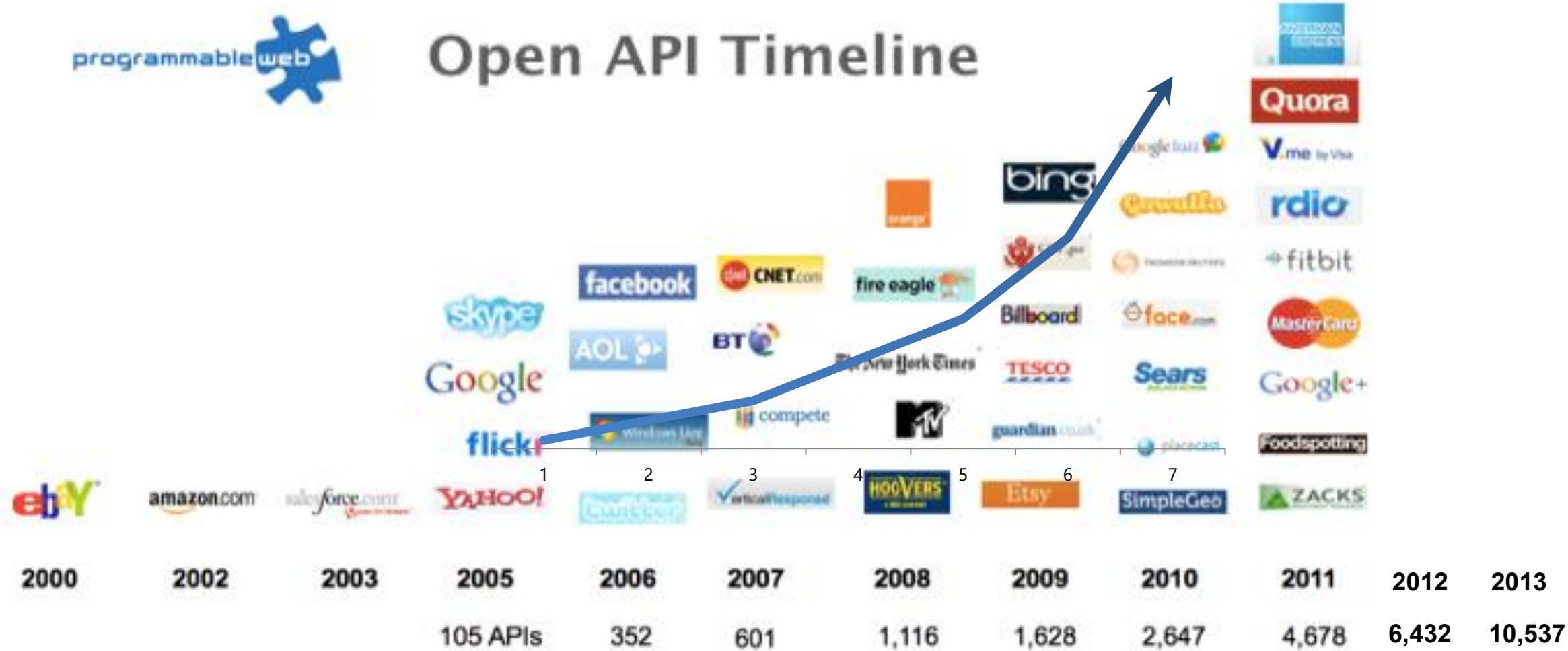
Data

...and 5000+
people work on
tooling for this

Proposition 1

We are living through an information
revolution

The Information Revolution



Proposition 2

Our programming languages are
information-sparse

Proposition 3

This is a big problem

for all languages
(Java, C#, F#, Python, C++, ...)

We need to bring information into the
language...

At internet-scale, strongly tooled, strongly typed

But before we get into that...

A bit about F#

F# is a
programming
language

F# is a companion
to C#, widely used
in enterprises

C#, F# and .NET are
free, open, cross-
platform

Xamarin provide F#
tools for Android
and iOS

Microsoft provide
F# tools for
Windows/Azure in
Visual Studio

Debian and other
packages for Linux

The F# language is
accepting
contributions

fsharp.org

F# 4.0 just complete

Example #1 (power company)

We have written an application to balance the national power generation schedule ... for an energy company.

...the calculation engine was written in F#.

The use of F# to address the complexity at the heart of this application clearly demonstrates a sweet spot for the language ... algorithmic analysis of large data sets.

Simon Cousins (Eon Powergen)

Example #1 (power company)

Time to Market

Interoperation ... Seamless. The C# programmer need never know.

Parallelism ...The functional purity ... makes it ripe for exploiting the inherent parallelism in processing vectors of data.

Efficiency

Units of measure ... a huge time saver...it eradicates a whole class of errors

Correctness

Code reduction... ... vector matrices...higher order functions eat these for breakfast with minimal fuss, minimal code. Beautiful.

Time to Market

Exploratory programming ... Working with F# Interactive allowed me to explore the solution space more effectively

Time to Market

Unit testing ...a joy to test. There are no complex time-dependent interactions to screw things up....

Correctness

Lack of bugs... Functional programming feel strange. ... once the type checker is satisfied that's often it, it works.

Correctness

Example #1 (Simon Cousins, Energy Sector)

350,000

lines of C# OO
by offshore team

The C# project took five years and peaked at ~8 devs. It never fully implemented all of the contracts.

The F# project took less than a year and peaked at three devs (only one had prior experience with F#). All of the contracts were fully implemented.

30,000

lines of robust F#, with
parallel + more features

An application to evaluate the revenue due from [Balancing Services](#) contracts in the UK energy industry

<http://simontcousins.azurewebsites.net/does-the-language-you-use-make-a-difference-revisited/>

Example #1 (Simon Cousins, Energy Sector)

Zero

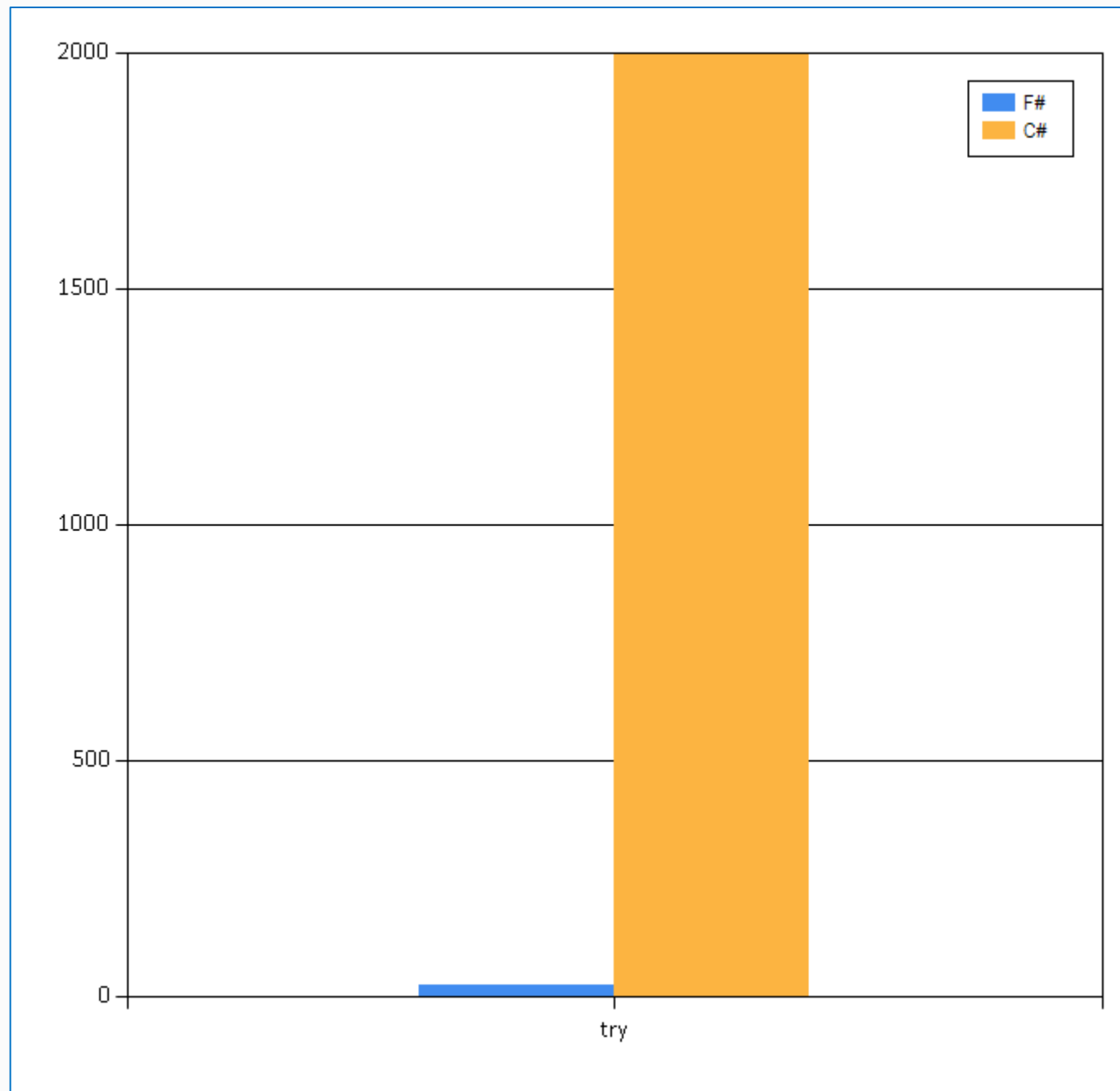
bugs in deployed system

“F# is the safe choice for this project,
any other choice is too risky”

An application to evaluate the revenue due from [Balancing Services](#) contracts in the UK energy industry

<http://simontcousins.azurewebsites.net/does-the-language-you-use-make-a-difference-revisited/>

Implementation	C#	F#
Braces	56,929	643
Blanks	29,080	3,630
Null Checks	3,011	15
Comments	53,270	487
Useful Code	163,276	16,667
App Code	305,566	21,442
Test Code	42,864	9,359
Total Code	348,430	30,801



Example #2: F# for Cloud Social Gaming

F# is becoming an increasingly important part of our server architecture that supports our mobile and web-based social games with millions of active users. F# first came to prominence in our technology stack in the implementation of the rules engine for our social slots games which by now serve over 700,000 unique players and 150,000,000 requests per day at peaks of several thousand requests per second.

Efficiency

The F# solution offers us an order of magnitude increase in productivity and allows one developer to perform the work that are performed by 10 dedicated developers on an existing Java-based solution, and supporting our agile approach and bi-weekly release cycles.

Time to Market

Yan Cui, Lead Server Engineer
fsharp.org/testimonials

F# for Android, iOS, Linux, Mac

fsharp.org/use/android

fsharp.org/use/ios

fsharp.org/use/linux

fsharp.org/use/mac

Programming + Data

Data is like water...



Data is like water...

- Everyone needs it. Everyone knows where to get it.
- Nobody is sure where it really came from, or goes to.
- ...really knows its true cost, or true value.
- ...likes to pay for it, or to share it.
- ...knows how much is wasted.
- You might get washed away by it.
- You only find out it was bad after you have drunk it.

Actually these days it's more like a flood...



The Problem

Our programming tools are **data-sparse**

getting data **into** a programming language is tiresome, error prone and boring

The Problem

Our data tools are programming-sparse

doing programming in a data tool is tiresome, error prone, weak

We need to bring data into the
language...

At internet scale, strongly tooled, strongly typed

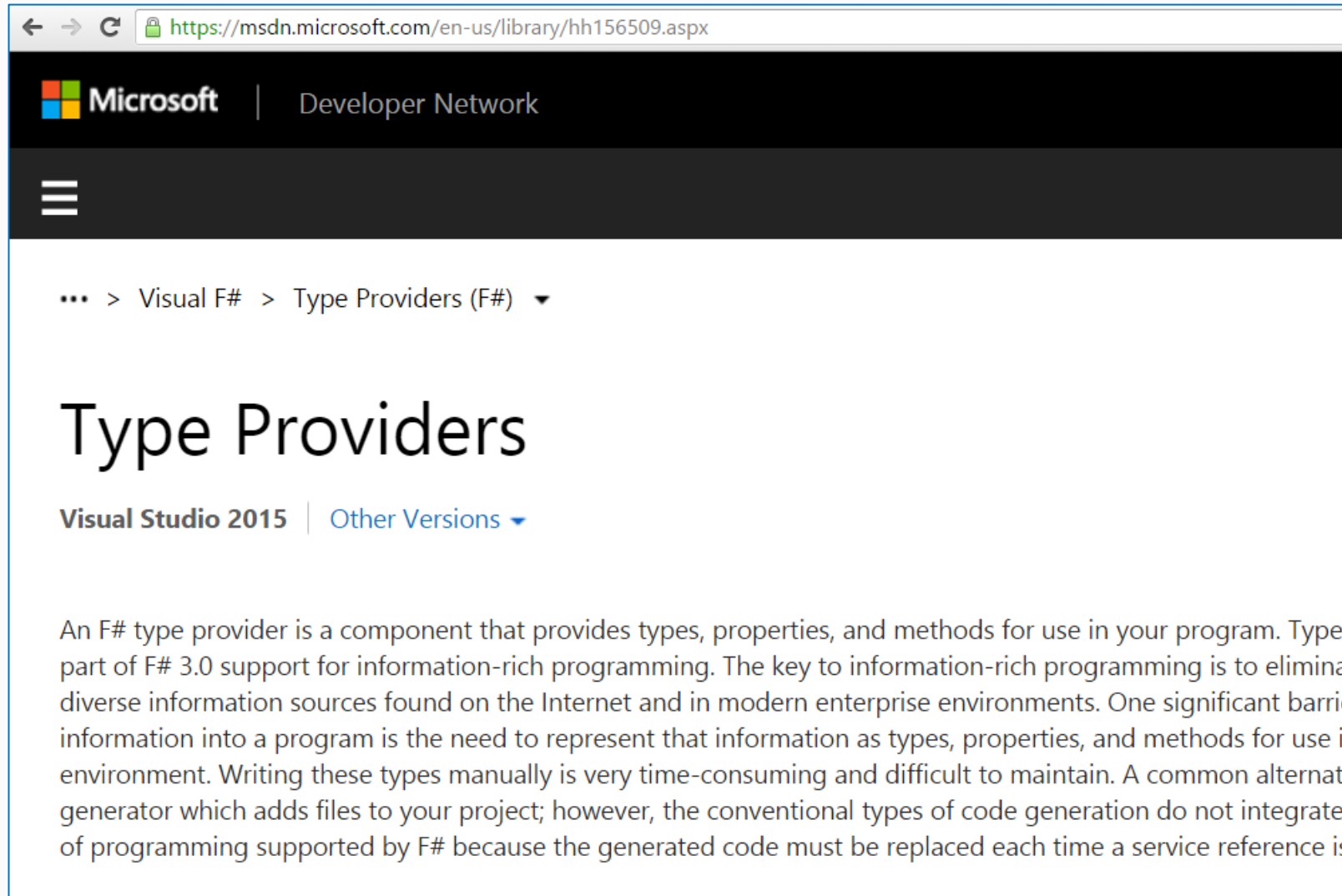
Problem: Integrate all of
worldbank.org/data
"as if it were a library"

Demo

F# + WorldBank

An F# type provider for a specific large data source

Enter F# Type Providers...



The screenshot shows a web browser window with the URL <https://msdn.microsoft.com/en-us/library/hh156509.aspx>. The page header features the Microsoft logo and "Developer Network". A breadcrumb trail reads "... > Visual F# > Type Providers (F#)". The main heading is "Type Providers". Below it, there are links for "Visual Studio 2015" and "Other Versions". The introductory text explains that an F# type provider is a component that provides types, properties, and methods for use in a program, highlighting its role in information-rich programming and its integration with F# 3.0.

← → ↻ <https://msdn.microsoft.com/en-us/library/hh156509.aspx>

 Microsoft | Developer Network

☰

... > Visual F# > Type Providers (F#) ▾

Type Providers

Visual Studio 2015 | [Other Versions](#) ▾

An F# type provider is a component that provides types, properties, and methods for use in your program. Type part of F# 3.0 support for information-rich programming. The key to information-rich programming is to eliminate diverse information sources found on the Internet and in modern enterprise environments. One significant barrier to integrating information into a program is the need to represent that information as types, properties, and methods for use in the programming environment. Writing these types manually is very time-consuming and difficult to maintain. A common alternative is to use a code generator which adds files to your project; however, the conventional types of code generation do not integrate with the advanced features of programming supported by F# because the generated code must be replaced each time a service reference is updated.

Theme #1 - Many Data Sources, One Mechanism



CATS: ALL YOUR types ARE BELONG
TO US.

SQL #1

```
type NorthwndDb =  
    SqlConnection<ConnectionString = @"AttachDBFileName = 'C:\project:  
  
let db = NorthwndDb.GetDataContext()  
  
let customerNames =  
    query { for c in db. do  
        where (c.Ci  
        select c.Con  
        AlphabeticalListOfProducts  
        Categories  
        CategorySalesFor1997s  
        property  
        NorthwndDb.ServiceTypes.Simple  
        phabeticalListOfProducts:  
        System.Data.Linq.Table<Northwnd
```

SQL #2

```
let connectionString = @"Data Source=(LocalDb)\v11.0;Initial Catalog=Adventureworks2012"

[<Literal>]
let query = "
    SELECT TOP(@TopN) FirstName, LastName, SalesYTD
    FROM Sales.vSalesPerson
    WHERE CountryRegionName = @regionName AND SalesYTD > @salesMoreThan
    ORDER BY SalesYTD
"

type SalesPersonQuery = SqlCommandProvider<query, connectionString>
let cmd = SalesPersonQuery()
```

CSV

```
open FSharp.Data

type Stocks = CsvProvider<"C:\\data\\aapl.csv">
let fb = Stocks.Load("http://ichart.finance.yahoo.com/table.csv?s=FB")
```

```
for r in fb.Rows do
    if r.
```

- ⚙ Adj Close
- ⚙ Close
- ⚙ Date
- ⚙ High
- ⚙ Low
- ⚙ Open
- ⚙ Volume

JSON

```
open FSharp.Data
```

```
[<Literal>]
```

```
let sample = "http://api.openweathermap.org/data/2.5/weather?q=London"
```

```
let apiUrl = "http://api.openweathermap.org/data/2.5/weather?q="
```

```
type Weather = JsonProvider<sample>
```

```
let sf = Weather.Load(apiUrl + "San Francisco")
```

```
sf.Sys
```

-  Country property JsonProvider<...>.Sys.Country: string
-  JsonValue
-  Message
-  Sunrise
-  Sunset

XML

```
1: type Author = XmlProvider<""<author name="Paul Feyerabend" born="1924" />"">
2: let sample = Author.Parse("""<author name="Karl Popper" born="1902" />""")
3:
4: printfn "%s (%d)" sample.Name sample.Born
```

Hadoop/Hive

```
type HadoopData = HiveTypeProvider<"tryfsharp",Port=10000,DefaultTimeo
```

```
let data = HadoopData.GetDataContext()
```

```
let testQuery1 =
```

```
    query { for x in data. do  
        select x }
```

```
module AbaloneCatchAnalysi
```

- ExecuteQuery
- GetTable
- GetTableMetadata
- GetTableNames
- Host
- Port
- UserName
- abalone

00 %

Interactive

Web APIs

```
#r "../TypeProviders/Debug/net40/Samples.WorldBank.dll"
```

```
let data = Samples.WorldBank.GetDataContext()
```

```
data.Countries.
```

```
data.Countries.
```

- ✎ Afghanistan
- ✎ Albania
- ✎ Algeria
- ✎ American Samoa
- ✎ Andorra
- ✎ Angola
- ✎ Antigua and Barbuda
- ✎ Arab World

-14 (% of total)

0 %

Interactive

Entity Graphs

```
#r @"..\TypeProviders\Debug\net40\Samples.DataStore.Freebase.dll"
```

```
open Samples.DataStore.Freebase
```

```
// Access the service types using our API key
```

```
type Freebase = FreebaseDataProvider<Key=API_KEY>
```

```
let ctxt = Freebase.GetDataContext()
```

```
ctxt.``Arts and Entertainment``.
```

- ✎ Books
- ✎ Broadcast
- ✎ Comics
- ✎ Fictional Universes
- ✎ Film
- ✎ Games
- ✎ Media
- ✎ Music

property

FreebaseDataProvider<...>.ServiceTypes.Dor

Entertainment.Books:

FreebaseDataProvider<...>.ServiceTypes.Dor

main

The publishing domain is home to most aspects of the written word -- books, magazines, scholarly academic papers, etc. Most of the data we have imported from Wikipedia, although we are looking for other possible data sources. We encourage authors, writings, or publications if we're missing information, please see the documentation for

```
1 data : HiveTypeProvider<...>.DataTypes
```

WSDL

```
type TerraService = WsdService<"http://msrmaps.com/TerraService2.asmx?WSDL">  
  
let terraClient = TerraService.GetTerraServiceSoap ()  
    let myPlace = new TerraService.ServiceTypes.msrmaps.com.Place(City = "Redmond")  
    let myLocation = terraClient.ConvertPlaceToLonLatPt(myPlace)  
    printfn "Redmond Latitude: %f Longitude: %f" (myLocation.Lat) (myLocation.Longitude)
```

R

```
// Pull in stock prices for some tickers then compute returns
let data = [
  for ticker in [ "MSFT"; "AAPL"; "VXX"; "SPX"; "GLD" ] ->
    ticker, getStockPrices ticker 255 |> R.log |> R.diff ]

// Construct an R data.frame then plot pairs of returns
let df = R.data_frame(namedParams data)
R.pairs(df)
```

SQL #2 - Application

Tachyus is a Silicon Valley startup that aims to be “*a Data Start-Up for the Oil Industry*”. They aim to create an array of sensors and mobile applications to help oil and gas producers better record and analyze their wells. According to [the New York Times coverage](#):

The start-up represents an anomaly of sorts in Silicon Valley. Many new businesses focus on high-technology products for the Internet or green technology, but Mr. Sloss and his co-founders, Paul Orland and Francisco LePort, have instead homed in on the decidedly older and dirtier business of drilling for hydrocarbons.



Last week Tachyus announced that it has raised \$6M in funding from a group led by Founders Fund. At the time of the announcements, one of the Tachyus engineers announced that they went from “from zero to product launch in 12 weeks” and “we couldn’t have done it without F#”. Founder Paul Orland commented “we are using 100% F#”



Retweeted by [Community for F#](#)

Jack Fox @foxyjackfox · Apr 1

#tachyus went from 0 to product launch today in 12 weeks. Could not have done it without #fsharp tachyus.com

Collapse

↩ Reply ↻ Retweet ★ Favorite ... More

Theme #2 - Data at Multiple Scales

From Everything to Individuals

```
data.AllEntites
```

```
data.Automotive.``Automobile Models``
```

```
data.Automotive.``Automobile Models``.Individuals.``Porsche 911``
```

Data Scripters need to work with different granularities of schematization

...Only a language that supports massively scalable metadata can operate at all these levels

Every stable entity can get a unique type

*"The idea of **integrating internet-scale information services** directly into the program's variable and type space is **probably the most innovative programming language idea** I've heard of in a decade"*

(Jim Cordy, Queens U)

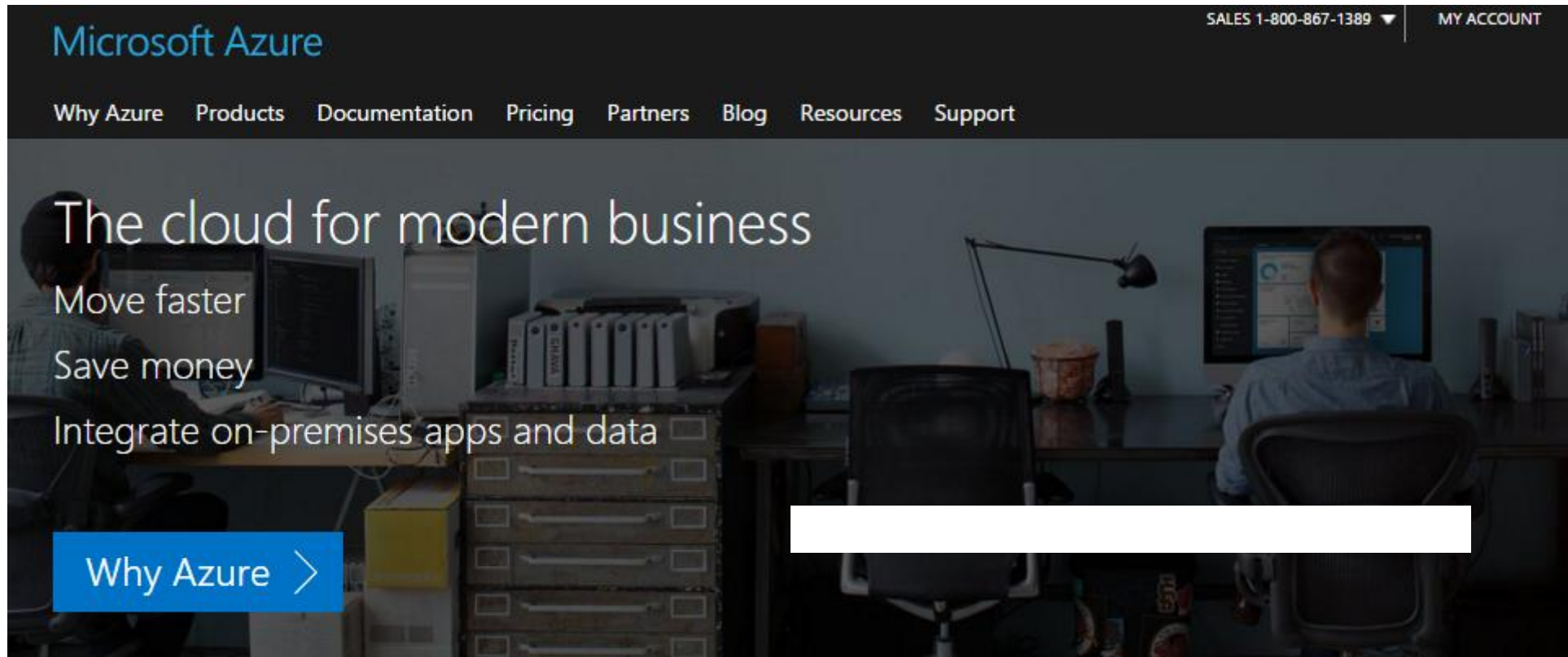
Overly kind, but thought provoking!

Programming + Data +
Cloud

The problem: cloud programming is hard

The solution: simple, scalable cloud+data
scripting

Context: Microsoft Azure



Microsoft Azure

SALES 1-800-867-1389 ▼ MY ACCOUNT

Why Azure Products Documentation Pricing Partners Blog Resources Support

The cloud for modern business

Move faster
Save money
Integrate on-premises apps and data

Why Azure >

[Redacted]

Sign up and deploy your first cloud solution in under 5 minutes



Welcome to **simple scripting of scalable compute and data jobs**

MBrace.Core is a simple programming model for scalable cloud data scripting and programming with F# and C#. With MBrace.Azure, you can script Azure for large-scale compute and data processing, directly from your favourite editor.

<http://mbrace.io/>

100% open source, community driven

Summary

Summary

At MSR we have deep
expertise in
programming tech

Data + Cloud +
Programming is a
major challenge

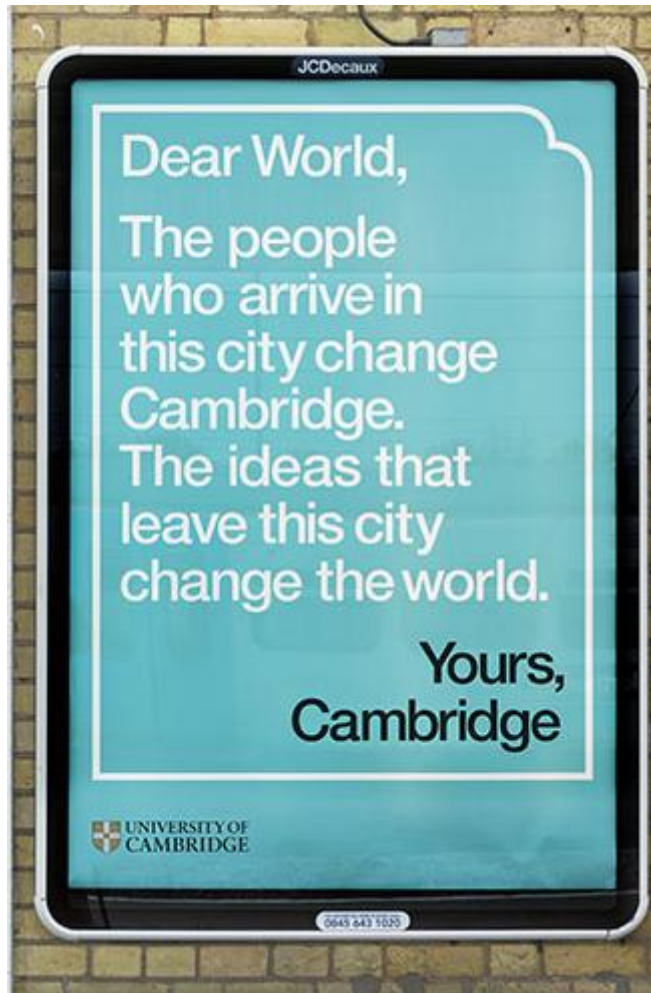
Correct, scalable
programming lies at
the heart of successful
tech companies

F# is an enterprise-
ready tool uniquely
suited to data + cloud
+ programming

Questions?



Ideas matter!



To: Apple, Microsoft,
Lenovo, Intel,
Dell, Acer, Sony,
IBM, LG

cc: Google, Facebook,
Yahoo, AOL, Twitter,
YouTube, eBay,
Tumblr, Pinterest,
Amazon

Glad things
are going well.
Yours, Charles
and Alan

Charles Babbage (Trinity College 1810, Peterhouse 1812), inventor and mathematician
Alan Turing (King's College 1931), computer scientist and cryptographer
cam.ac.uk/YoursCambridge

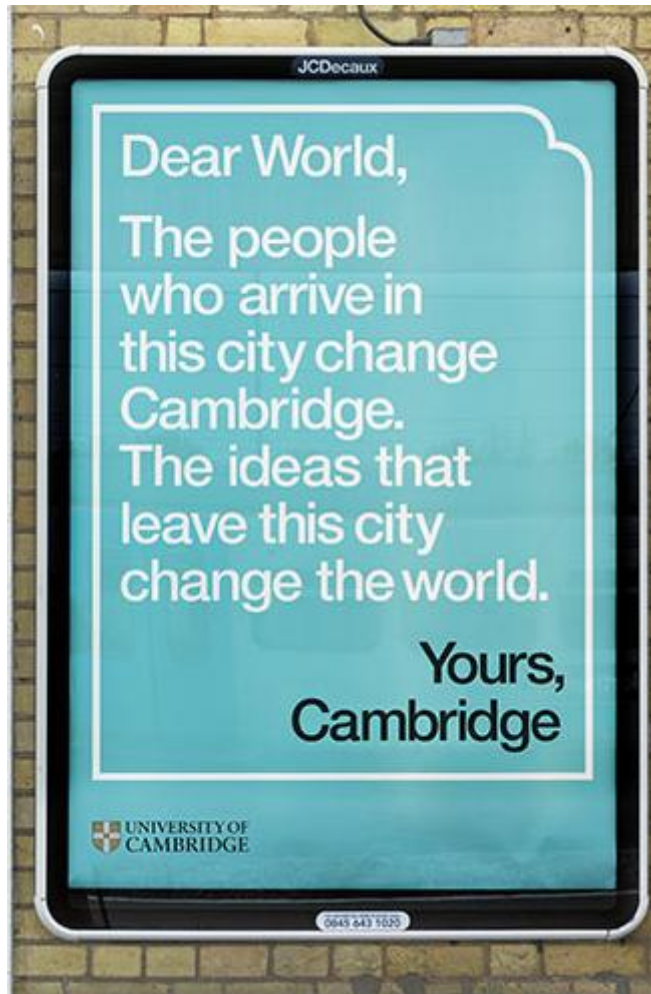
Dear Quentin,
Black, no sugar.



Yours, Paul

Quentin Stafford-Fraser (Gonville & Caius College 1986) and Paul Jardetsky (Darwin College 1988), makers of the first webcam in 1991, to keep an eye on the coffee pot cam.ac.uk/YoursCambridge

Ideas matter!



To: Apple, Microsoft,
Lenovo, Intel,
Dell, Acer, Sony,
IBM, LG

cc: Google, Facebook,
Yahoo, AOL, Twitter,
YouTube, eBay,
Tumblr, Pinterest,
Amazon

Glad things
are going well.
Yours, Charles
and Alan

Charles Babbage (Trinity College 1810, Peterhouse 1812), inventor and mathematician
Alan Turing (King's College 1931), computer scientist and cryptographer
cam.ac.uk/YoursCambridge

To: developers

Cup<T>?
async { ? }
(|A|B|) ?
float<kg>?
(a,b)?
let x = ...

Yours,
Microsoft Research
Cambridge

Providing Units of Measure

via F#'s Units of Measure

If the metadata contains units...

Dissipated	/meteorology/tropical_cyclone/dissipated	/type/datetime
Highest winds	/meteorology/tropical_cyclone/highest_winds	/type/float <i>Kilometres per hour</i>
Lowest Pressure	/meteorology/tropical_cyclone/lowest_pressure	/type/float <i>Millibar</i>
Damages	/meteorology/tropical_cyclone/damages	/measurement_unit/dated_money
Di		

```
let cyclones = data.`Science and Technology`.Meteorology.`Tropical
```

```
let topWind = cyclones.`Hurricane $`
```

```
val topWind : float<metre/second>
```

Full name: Demo.topWind

...then these can be projected into the programming language.

Aside: Excel!!

